

Voltage Controlled Oscillator

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Exact and Approximate Propagation Delay Analysis of a CMOS Inverter

1 CMOS Propagation Delay and Ring Oscillator VCO

1.1 Propagation Delay of CMOS Inverter

The propagation delay of a CMOS inverter is defined as the average of the high-to-low and low-to-high delays:

2 Introduction

This report presents both the exact and approximate derivation of propagation delay in a CMOS inverter. The exact model uses MOSFET current equations, while the approximate model simplifies the inverter using an RC equivalent.

3 Basic Equation

The output node is modeled as a load capacitance C_L . The governing equation is:

$$C_L \frac{dV_{out}}{dt} = \pm I_D \quad (1)$$

4 High-to-Low Delay (t_{pHL})

When the input goes HIGH:

- NMOS turns ON
- PMOS turns OFF
- Output discharges from V_{DD} to 0

4.1 Saturation Region

$$V_{out} \geq V_{DD} - V_{tn} \quad (2)$$

$$I_{Dn} = \frac{1}{2}k_n(V_{DD} - V_{tn})^2 \quad (3)$$

$$t_{sat,n} = \frac{2C_L V_{tn}}{k_n(V_{DD} - V_{tn})^2} \quad (4)$$

4.2 Triode Region

$$V_{out} < V_{DD} - V_{tn} \quad (5)$$

$$I_{Dn} = k_n \left[(V_{DD} - V_{tn})V_{out} - \frac{V_{out}^2}{2} \right] \quad (6)$$

$$t_{triode,n} = \frac{C_L}{k_n(V_{DD} - V_{tn})} \ln \left(\frac{2(V_{DD} - V_{tn}) - V_{out}}{V_{out}} \right) \bigg|_{V_{DD}-V_{tn}}^{V_{DD}/2} \quad (7)$$

$$t_{pHL} = t_{sat,n} + t_{triode,n} \quad (8)$$

5 Low-to-High Delay (t_{pLH})

When the input goes LOW:

- PMOS turns ON
- NMOS turns OFF
- Output charges from 0 to V_{DD}

5.1 Saturation Region

$$I_{Dp} = \frac{1}{2}k_p(V_{DD} - |V_{tp}|)^2 \quad (9)$$

$$t_{sat,p} = \frac{2C_L |V_{tp}|}{k_p(V_{DD} - |V_{tp}|)^2} \quad (10)$$

5.2 Triode Region

$$I_{Dp} = k_p \left[(V_{DD} - |V_{tp}|)(V_{DD} - V_{out}) - \frac{(V_{DD} - V_{out})^2}{2} \right] \quad (11)$$

$$t_{triode,p} = \frac{C_L}{k_p(V_{DD} - |V_{tp}|)} \ln \left(\frac{2(V_{DD} - |V_{tp}|) - (V_{DD} - V_{out})}{V_{DD} - V_{out}} \right) \quad (12)$$

$$t_{pLH} = t_{sat,p} + t_{triode,p} \quad (13)$$

6 Approximate Delay Model (RC Approximation)

6.1 Assumptions

- MOSFETs are modeled as linear resistors
- Load is purely capacitive (C_L)
- Input is a step signal
- Short-circuit current is neglected
- Average current approximation is used

7 Derivation of Equivalent Resistances

7.1 NMOS Resistance (R_n)

When NMOS is ON:

$$I_{Dn} \approx \frac{1}{2}k_n(V_{DD} - V_{tn})^2 \quad (14)$$

Effective resistance is defined as:

$$R_n = \frac{V_{DD}}{I_{Dn}} \quad (15)$$

Substituting:

$$R_n = \frac{V_{DD}}{\frac{1}{2}k_n(V_{DD} - V_{tn})^2} \quad (16)$$

$$R_n = \frac{2V_{DD}}{k_n(V_{DD} - V_{tn})^2} \quad (17)$$

7.2 PMOS Resistance (R_p)

Similarly for PMOS:

$$I_{Dp} \approx \frac{1}{2}k_p(V_{DD} - |V_{tp}|)^2 \quad (18)$$

$$R_p = \frac{V_{DD}}{I_{Dp}} \quad (19)$$

$$R_p = \frac{2V_{DD}}{k_p(V_{DD} - |V_{tp}|)^2} \quad (20)$$

8 Approximate Propagation Delay

8.1 High-to-Low Delay

$$t_{pHL} = 0.69R_nC_L \quad (21)$$

8.2 Low-to-High Delay

$$t_{pLH} = 0.69R_pC_L \quad (22)$$

8.3 Average Delay

$$t_p = \frac{t_{pHL} + t_{pLH}}{2} \quad (23)$$

$$t_p = 0.69C_L \frac{(R_n + R_p)}{2} \quad (24)$$

$$t_p = \frac{t_{pHL} + t_{pLH}}{2} \quad (25)$$

Using the RC approximation, the delay can be expressed as:

$$t_p \approx 0.69C_L \frac{(R_n + R_p)}{2} \quad (26)$$

where:

- C_L is the load capacitance
- R_n and R_p are the effective resistances of NMOS and PMOS

Thus, the propagation delay depends on device parameters and supply voltage.

8.4 Ring Oscillator using CMOS Inverters

A ring oscillator is formed by connecting an odd number (N) of CMOS inverters in a closed loop. Due to the odd number of inversions, the circuit cannot reach a stable operating point and oscillates continuously.

8.5 Oscillation Mechanism

A transition propagates through each inverter, and each stage introduces a delay of t_p . After passing through all stages, the signal is inverted and fed back, sustaining oscillation.

One complete oscillation requires the signal to propagate twice through the loop.

8.6 Time Period of Oscillation

The time period of oscillation is given by:

$$T = 2Nt_p \quad (27)$$

For a 5-stage ring oscillator ($N = 5$):

$$T = 10t_p \quad (28)$$

8.7 Frequency of Oscillation

The oscillation frequency is:

$$f = \frac{1}{T} = \frac{1}{2Nt_p} \quad (29)$$

For $N = 5$:

$$f = \frac{1}{10t_p} \quad (30)$$

8.8 Frequency in Terms of Circuit Parameters

Substituting the approximate delay:

$$f = \frac{1}{2N \cdot 0.69C_L \frac{(R_n + R_p)}{2}} \quad (31)$$

8.9 Voltage Controlled Oscillator Behavior

The oscillation frequency depends on the propagation delay, which is influenced by:

- Supply voltage V_{DD}
- Load capacitance C_L
- Transistor parameters (k_n , k_p)

Since:

$$f \propto \frac{1}{t_p} \quad (32)$$

any variation in control voltage changes the delay and hence the oscillation frequency. Therefore, the ring oscillator behaves as a Voltage Controlled Oscillator (VCO).

8.10 Design Insights

- Increasing C_L increases delay and reduces frequency
- Increasing transistor width reduces R_n and R_p , increasing frequency
- Increasing V_{DD} reduces delay and increases frequency

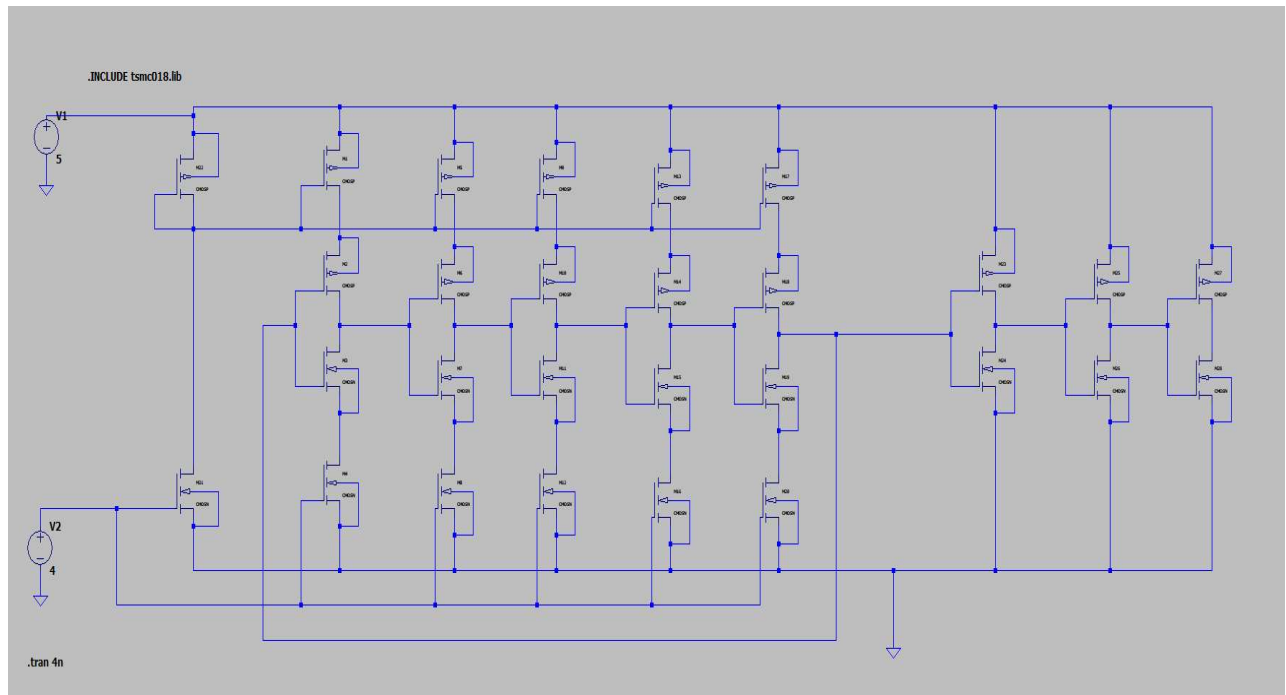
8.11 Condition for Oscillation

For sustained oscillations:

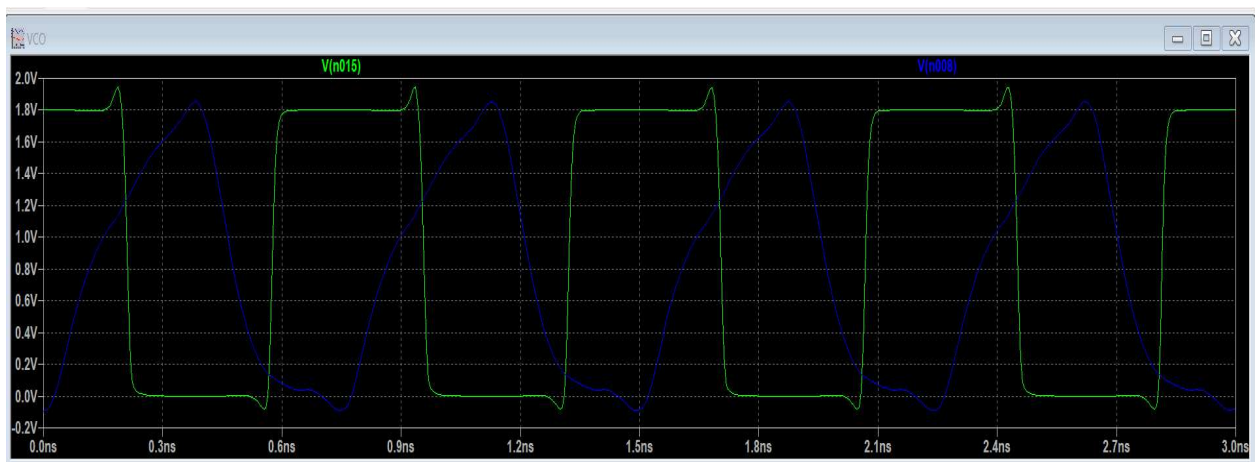
$$N \text{ must be odd} \quad (33)$$

An even number of stages results in a stable output and no oscillation.

Circuit Diagram



Results



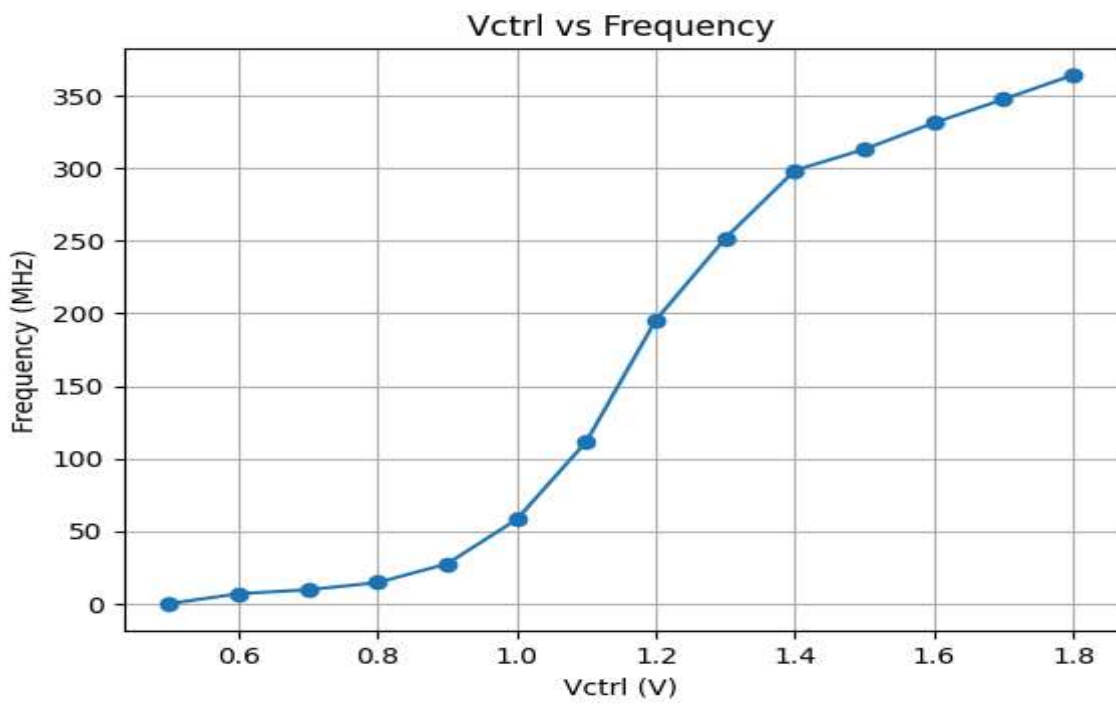
Output waveform (at Vctrl at 1V)



Plot of o/p waveforms by varying control voltages from 0.5V to 1.8V

Vctrl	Freq (MHz)
0.5	0
0.6	6.91
0.7	9.59
0.8	14.57
0.9	27.44
1.0	58.00
1.1	111.47

1.2	195.22
1.3	251.66
1.4	298.42
1.5	313.00
1.6	331.14
1.7	347.40
1.8	364.14



From above data and plot:

- Frequency increases from ~ 6.9 MHz $\rightarrow$ ~ 364 MHz
- As V_{ctrl} increases from 0.5 V $\rightarrow$ 1.8 V

Trend:

- **Frequency increases nonlinearly with V_{ctrl}**

Early region (0.5–0.8 V):

- Slow increase
- Large delay $\rightarrow$ weak current

Mid region (0.8–1.2 V):

- Rapid increase
- Strong inversion begins

High region (1.2–1.8 V):

- Saturation behavior
- Frequency increase slows slightly

Best linear region is roughly:

$V_{ctrl} \approx 0.9\text{V to } 1.3\text{V}$